

-XX:[+|-]FlightRecorder

 This option enables or disables JDK Flight Recorder (JFR) in the VM. This built-in profiling and troubleshooting feature in the VM collects profiling and diagnostic information.

:fontawesome-solid-pencil:{: .note aria-hidden="true"} **Notes:**

- OpenJ9 offers limited support for JFR functionalities.
- JFR recording can be enabled or disabled on OpenJ9 by using either the `jcmd` options as specified in the Explanation section or the `-XX:StartFlightRecording` command-line option.
- The JFR-related `jcmd` options cannot be used simultaneously with the command-line option.
- The JDK Mission Control (JMC) tool is the only way to view the produced JFR file.

Syntax

```
-XX:[+|-]FlightRecorder
```

Setting	Effect	Default
<code>-XX:+FlightRecorder</code>	Enable	:fontawesome-solid-check:{: .yes aria-hidden="true"}yes
<code>-XX:-FlightRecorder</code>	Disable	

Explanation

If JFR is enabled in the VM with the `-XX:+FlightRecorder` option, then you can trigger profile and diagnostic recording with either the `jcmd` tool or the `-XX:StartFlightRecording` command-line option. Recording does not start automatically, it must be triggered.

To start a recording by using the `jcmd` options, specify the following `jcmd` option command:

```
jcmd <vmid | display name | 0> JFR.start [filename=<file_name_with_path>]  
[duration=<time_with_unit_of_time>]
```

where:

- Optional parameters are shown in brackets, [].
- `filename=<file_name_with_path>` specifies the name of the file and its location where the recording is saved. The file name should have a `.jfr` extension. If a file name is not specified, the recording is saved in the `defaultJ9recording.jfr` file in the process working directory by default.

- `duration=<time_with_unit_of_time>` specifies the duration of the recording. Units of time are `s` (seconds), `m` (minutes), `h` (hours), and `d` (days). You can specify only one unit of time, for example, `54s`, `12m`, `1h`, or `2d`. If the duration is not specified, the tool continues to record until the recording is manually stopped or the VM is shut down.

Example:

```
jcmd 1234 JFR.start filename=/path/ABCD.jfr duration=4s
```

To stop a recording, specify the following command:

```
jcmd <vmid | display name | 0> JFR.stop
```

All JFR-related `jcmd` options might change in future releases.

Limitations:

- JFR produces only the following events:

- ClassLoadingStatistics
- CPUInformation
- CPULoad (except z/OS)
- ExecutionSample
- GarbageCollection
- GCHeapConfig
- GCHeapSummary
- InitialEnvironmentVariable
- InitialSystemProperty
- JVMInformation
- ModuleExports
- ModuleRequires
- MonitorEnter
- MonitorWait
- NativeLibrary (except z/OS)
- NetworkUtilization
- OldGarbageCollection
- OSInformation
- PhysicalMemory
- SystemGC
- SystemProcess (except z/OS)
- ThreadContextSwitchRate
- ThreadCPULoad (except z/OS)
- ThreadDump
- ThreadEnd
- ThreadPark
- ThreadSleep

- ThreadStart
- ThreadStatistics
- VirtualizationInformation
- YoungGarbageCollection
- YoungGenerationConfig

:fontawesome-solid-pencil:{: .note aria-hidden="true"} **Note:** To record and analyze the SystemProcess JFR event on z/OS (2.5 and 3.1), you must install [APAR OA62757](#).

- Support for the `-XX:StartFlightRecording` capability was made available in the 0.59.0 release.
- Support for the following capabilities is not available:
 - `-XX:FlightRecorderOptions`
 - JFR APIs (`jdk/jfr/*classes`)
 - Controlling JFR with Java Management Extensions (JMX)
 - JFR streaming
 - The `jfr` command-line utility

Support for JFR will be expanded in future releases.



See Also

- [What's new in version 0.49.0](#)
- [What's new in version 0.51.0](#)
- [What's new in version 0.53.0](#)
- [What's new in version 0.54.0](#)
- [What's new in version 0.56.0](#)
- [What's new in version 0.59.0](#)
- [-XX:StartFlightRecording](#)